Global Fuel Prices and Taxation Trends: Project Report

Oct 28, 2026 · @Vidhi Sharma

1. INTRODUCTION

1.1 Project Overview

Fuel prices differ greatly from one country to another, and so do the taxes and subsidies behind them. This project analyses weekly fuel price and tax data for 84 countries from January 2020 to April 2026, a total of 27,468 records. The data covers petrol, diesel and LPG prices in US dollars per litre, the Brent crude oil price, the tax percentage, and each country's region, income level and subsidy level.

The analysis is built in Tableau and presented as one interactive dashboard of seven linked charts: a world map of average petrol prices, the top 10 most expensive countries, the price trend over time, a comparison of petrol with Brent crude, average tax share by region, and average prices by income level and by subsidy level.

1.2 Purpose

The purpose of the project is to turn a large weekly dataset into a clear, visual explanation of how fuel is priced and taxed around the world. The dashboard is designed to:

- show which countries pay the most and the least for petrol;
- show how income level and subsidies relate to the pump price;
- show how much tax is included in the fuel price in each region;
- show how petrol and diesel prices have moved with crude oil since 2020.

2. IDEATION PHASE

2.1 Problem Statement

The raw dataset is a CSV file with more than 27,000 weekly rows. It is hard to read in a spreadsheet, and the patterns that matter (price gaps between countries, the effect of subsidies, the link to crude oil) cannot be seen without summarising and visualising the data.

I am	A student or analyst who compares fuel costs across countries
I am trying to	Understand why fuel prices differ and how they change over time
But	The data is a long weekly table, with noisy tax values, that I cannot read at a glance
Because	There is no single view that combines prices, tax, income, subsidy and crude oil

Which makes me feel	Unsure which factors really drive the pump price

2.2 Empathy Map Canvas

The empathy map describes the main user of the dashboard: a student, analyst or policy reader who wants quick answers about fuel prices.

Says	Thinks
"Which countries pay the most for petrol?" "Is it tax or subsidy that makes the difference?" "Do prices follow oil?"	The data is too big to scan by hand. A picture will show the pattern faster than a table. The averages must be trustworthy.
Does	**Feels**
Filters the data by region and income level. Compares countries and years. Looks for the highest and lowest values.	Confused by weekly ups and downs. Confident when the charts agree with the numbers. Interested when clicking a chart updates the rest.

Pains	Gains
Large raw data, noisy weekly tax values, no single overview	One dashboard, clear comparisons, filters and click-to-filter, headline numbers on top

2.3 Brainstorming

Several ideas were considered for the dashboard. Each was rated on value to the user and effort to build, and the ideas with high value and reasonable effort were selected.

Idea	Value	Effort	Decision
World map of average petrol price	High	Low	Selected
Top 10 most expensive countries	High	Low	Selected
Price trend for petrol and diesel	High	Low	Selected
Petrol against Brent crude (dual axis)	High	Medium	Selected
Average tax share by region	High	Low	Selected
Prices by income level and subsidy level	High	Low	Selected
Forecasting future prices	Medium	High	Future scope
Prices in local currency	Low	High	Not selected
Live price feed	Medium	High	Future scope

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

The journey follows a user from first opening the dashboard to sharing what they learned.

Stage	Goal	Action	Experience	Opportunity
Discover	Know what the dashboard covers	Reads the title and the headline numbers	Clear scope: 84 countries, 2020 to 2026	Short KPI row at the top
Explore	Find the most and least expensive countries	Looks at the map and the top 10 chart	Patterns are visible at once	Tooltips with exact values
Analyze	Understand why prices differ	Compares income, subsidy and tax charts	Subsidy stands out as the main driver	Filters by region, income and subsidy
Decide	Draw a conclusion	Checks the trend and the Brent comparison	Petrol follows crude oil closely	Notes that explain the data limits
Share	Present the findings	Exports or publishes the dashboard	Easy to show in a meeting	Publish to Tableau Public

3.2 Solution Requirement

Functional requirements

ID	Requirement
FR-1	Load the weekly fuel price CSV into Tableau and recognise Country as a geographic field
FR-2	Show average petrol price by country on a world map
FR-3	Rank the 10 most expensive countries
FR-4	Show petrol and diesel price trends by quarter
FR-5	Compare petrol price with Brent crude on a dual-axis chart
FR-6	Show average tax share by region and average price by income and subsidy level
FR-7	Provide filters (region, income, subsidy) and click-to-filter between charts

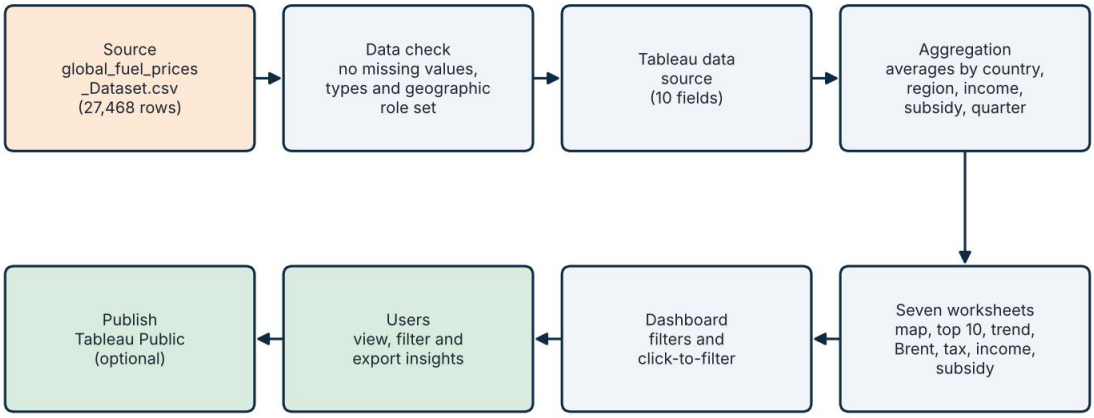
Non-functional requirements

ID	Requirement
NFR-1	Accuracy: every chart must use averages, and the values must match an independent calculation

ID	Requirement
NFR-2	Readability: consistent colours, titles and labels across all charts
NFR-3	Usability: a beginner can use the filters without instructions
NFR-4	Honesty: data limits are stated on the dashboard and in the report

3.3 Data Flow Diagram

Data moves from the CSV file through a data check and the Tableau data source, is aggregated into averages, feeds the seven worksheets, and is combined into the dashboard that the user views.



3.4 Technology Stack

Layer	Tool	Use in this project
Data	CSV file (global_fuel_prices_Dataset.csv)	Source of 27,468 weekly records
Visual analytics	Tableau (Desktop or Public)	Worksheets, dashboard, filters and actions
Map tiles	Mapbox and OpenStreetMap (inside Tableau)	Base map for the country map
Verification	Python with pandas	Independent check of the averages shown in Tableau
Reporting	PowerPoint and PDF	Presentation deck and this report

4. PROJECT DESIGN

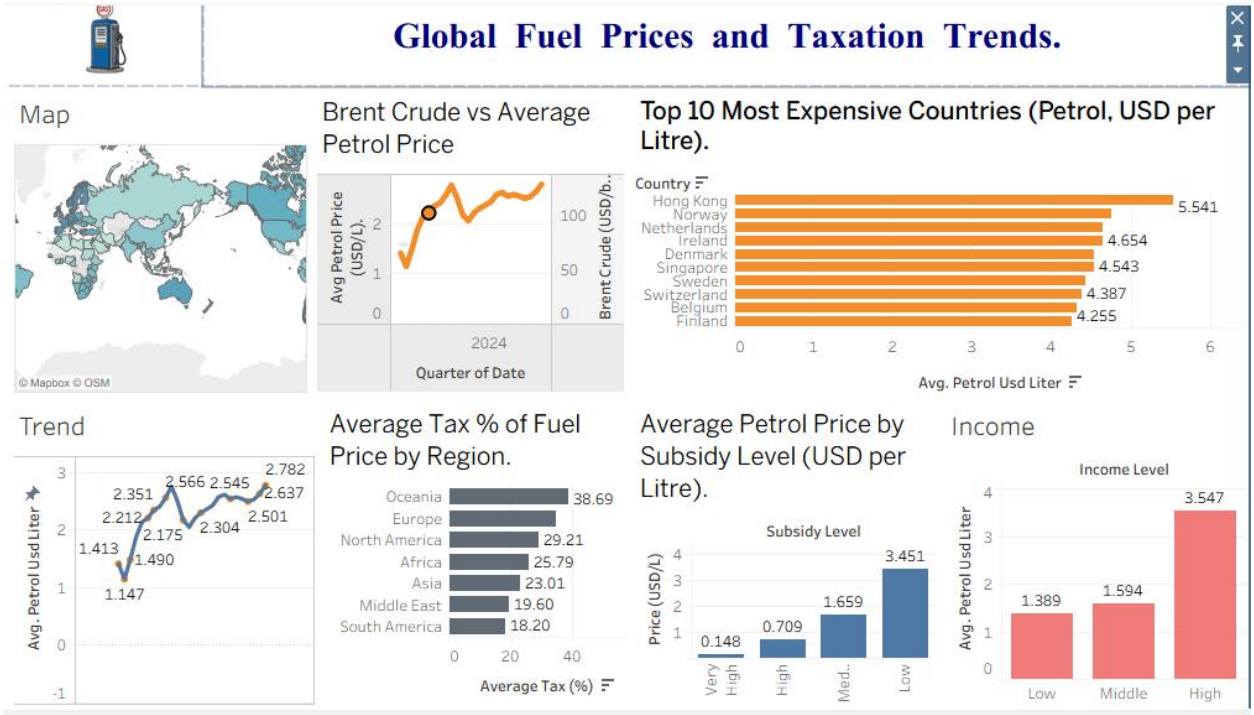
4.1 Problem Solution Fit

Problem	Solution in the dashboard	Fit

Problem	Solution in the dashboard	Fit
Large weekly table is hard to read	Averages shown in seven focused charts	Patterns visible at a glance
Cannot see which countries pay most	World map and top 10 chart	Highest and lowest countries found in seconds
Unclear what drives price	Income and subsidy charts side by side	Subsidy shown as the stronger factor
Tax values vary from week to week	Averages by region	Stable, comparable figures
Link to crude oil is unclear	Petrol against Brent on one chart	Close movement is obvious

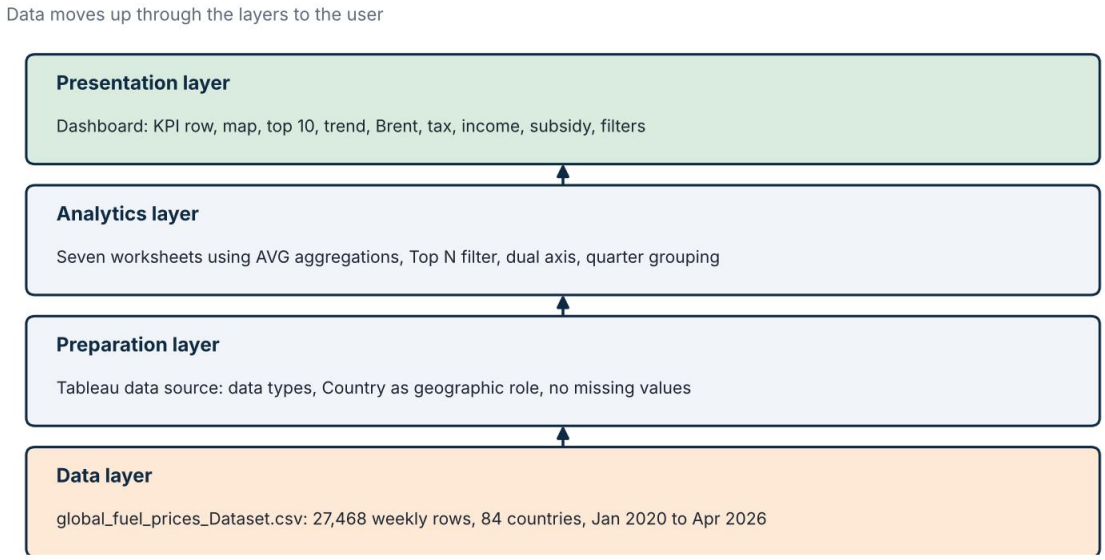
4.2 Proposed Solution

The proposed solution is a single interactive Tableau dashboard. A title and a row of headline numbers sit at the top. The world map and the top 10 chart show where fuel is expensive. The trend, Brent and tax charts show change over time and tax by region, and the income and subsidy charts explain differences between countries. Filters for region, income level and subsidy level are on the right, and clicking any chart filters the others.



4.3 Solution Architecture

The solution has four layers. The data layer holds the CSV file. The preparation layer defines data types and the geographic role in Tableau. The analytics layer computes averages and builds the seven worksheets. The presentation layer combines the worksheets into the dashboard for the user.



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activities	Date	Status
1. Understand the data	Read the dataset, check columns and missing values, define the questions, plan the sheets	26 Oct 2026	Completed
2. Build worksheets	Create the seven sheets, correct SUM to AVG, set quarterly dates, set the map as filled countries	27 Oct 2026	Completed
3. Test and correct	Check values against an independent calculation, remove leftover filters, fix axes and labels	27 to 28 Oct 2026	Completed
4. Presentation and report	Prepare the PowerPoint deck and this report	28 Oct 2026	Completed
5. Final dashboard polish	Align the dashboard layout, match colours and fonts	28 Oct 2026	In progress

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Accuracy was treated as the main performance measure for this project. Values shown in Tableau were compared with an independent calculation in Python (pandas) on the same CSV file, and every checked value matched.

Test	Expected result	Result
Data load	27,468 rows, 84 countries, no missing values	Passed
Average petrol price, Spain (map tooltip)	3.878 USD per litre	Passed (3.878)
Top 10 countries	Hong Kong 5.54, Norway 4.76, Netherlands 4.66 at the top	Passed
Income level averages	Low 1.39, Middle 1.59, High 3.55	Passed
Subsidy level averages	Very High 0.15, High 0.71, Medium 1.66, Low 3.45	Passed
Regional tax averages	Oceania 38.7 down to South America 18.2	Passed
Click-to-filter and filters	Selecting a country or a level updates the other charts	Passed

Testing also found defects during development. Each was corrected.

Defect found	Cause	Correction
Brent crude axis showed 800K and diesel reached 3,000	Measures were summed over all weeks and countries	Changed the aggregation to Average
Trend chart showed only Q1 to Q4	Quarter was used as a date part, merging all years	Used continuous quarterly dates
Map displayed as a list of countries	Country was on Rows, not latitude and longitude	Used generated latitude and longitude with Country on Detail
Charts went blank after clicking the dashboard	Leftover Action filters on the sheets	Removed the Action filters
Petrol and diesel drawn as two panels	Two separate measures on Rows	Merged into one dual-axis chart with a synchronised axis
Axis titles swapped on the Brent chart	Titles typed on the wrong axes	Left axis named for petrol, right axis for Brent

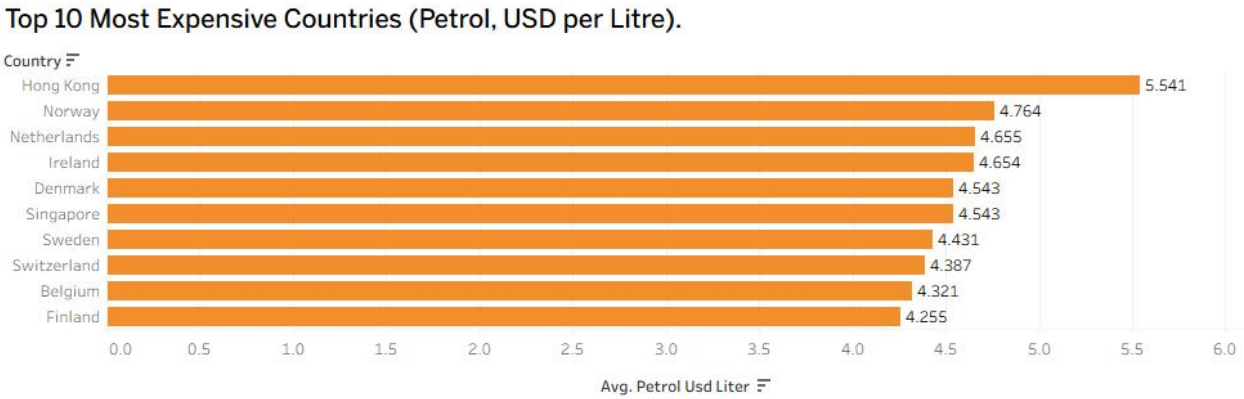
Response times were not formally measured. The dataset is about 1.8 MB, and no slow response was reported during development.

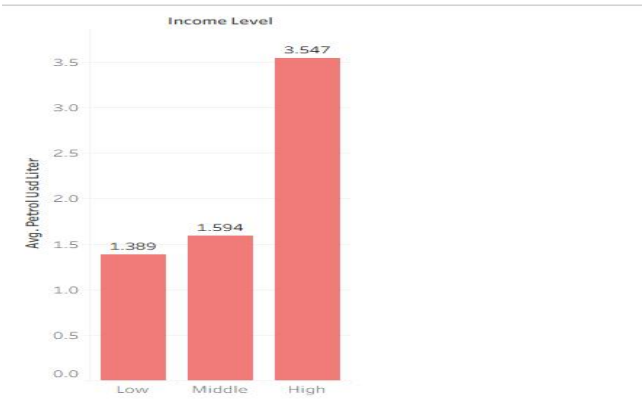
7. RESULTS

7.1 Output Screenshots

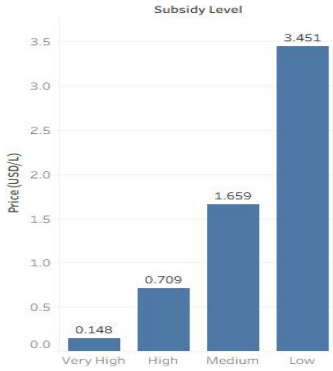
Figures 4 to 7 are screenshots taken from the Tableau workbook. Figures 8 and 9 were rebuilt from the same dataset to show the final quarterly view. You can replace any picture with your own final Tableau screenshot.

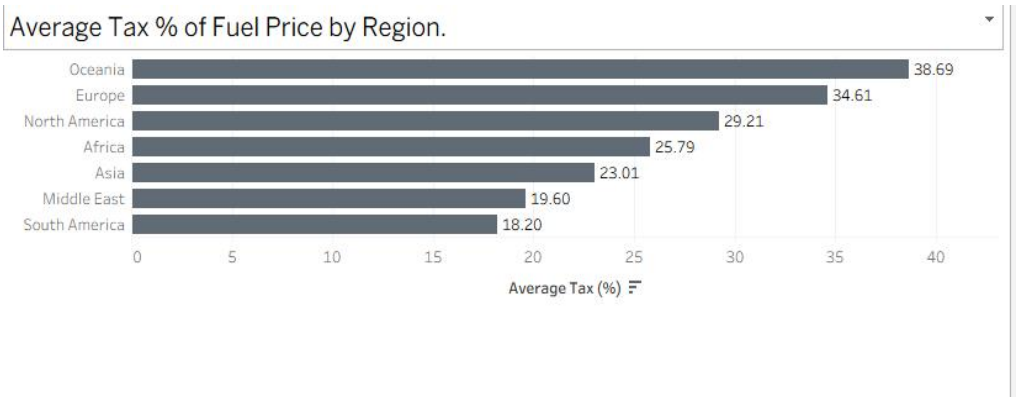




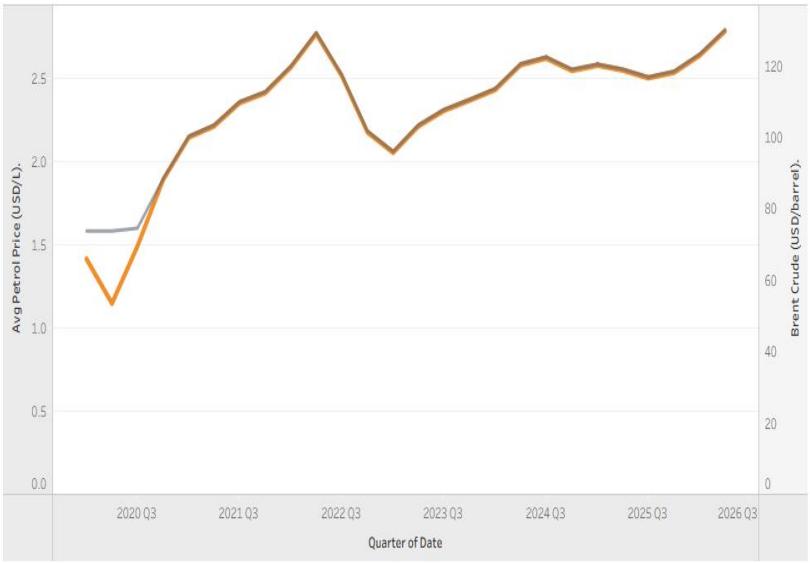


Average Petrol Price by Subsidy Level (USD per Litre).





Brent Crude vs Average Petrol Price



Key results

Finding	Result
Most and least expensive country (average petrol)	Hong Kong 5.54 and Venezuela 0.02 USD per litre
Effect of subsidy	Very High subsidy 0.15 against Low subsidy 3.45 USD per litre
Effect of income	Low income 1.39 against High income 3.55 USD per litre
Tax share by region	Oceania 38.7 percent, South America 18.2 percent
Petrol and Brent crude	Correlation of 0.99 on quarterly averages (2020 Q1 to 2026 Q1)
Highest quarterly petrol average	2.77 USD per litre in 2022 Q2

8. ADVANTAGES & DISADVANTAGES

Advantages

- One dashboard answers four clear questions about fuel prices.
- Interactive filters and click-to-filter let users explore without extra tools.
- All figures use averages and were checked against an independent calculation.
- The dashboard can be shared online through Tableau Public.

Disadvantages

- Weekly tax percentages vary widely for one country, so only averages can be trusted.
- The Brent crude value is the same for every country in a given week, so it shows the trend only.
- Regional and income averages hide differences between individual countries.
- The April 2026 data covers only one week of the second quarter, so the trend charts stop at Q1 2026.

9. CONCLUSION

This project turned 27,468 weekly records into an interactive Tableau dashboard that explains global fuel prices. The results show that subsidy level separates prices more sharply than income level (0.15 against 3.45 USD per litre for Very High and Low subsidy), that Oceania and Europe carry the highest tax shares, and that petrol prices follow Brent crude very closely (correlation 0.99). Hong Kong is the most expensive country and Venezuela the cheapest. Checking the values against an independent calculation showed that the dashboard is accurate, and the defects found during testing were corrected. The limits of the data are stated so that the findings are not over-read.

10. FUTURE SCOPE

- Add GDP per capita to test how income relates to tax and price.

- Add forecasting to estimate future petrol and diesel prices.
- Show tax in US dollars per litre as well as a percentage.
- Connect a live price source so the dashboard updates automatically.
- Add LPG price views, which are in the data but not yet charted.

11. APPENDIX

Source Code (if any)

The dashboard was built in Tableau without custom code. Python (pandas) was used only to check the averages shown in Tableau, for example:

```
import pandas as pd
df = pd.read_csv("global_fuel_prices_Dataset.csv")
print(df[df.country == "Spain"].petrol_usd_liter.mean())
print(df.groupby("income_level").petrol_usd_liter.mean())
print(df.groupby("subsidy_level").petrol_usd_liter.mean())
print(df.groupby("region").tax_percentage.mean())
```

Dataset Link

Dataset file: global_fuel_prices_Dataset.csv (27,468 rows, 10 columns, 6 January 2020 to 6 April 2026).

Online link:
https://drive.google.com/file/d/13k0qQW8OqvyIrE3yCzAym5SUTmQNd0qu/view?usp=sharing

GitHub & Project Demo Link

GitHub: https://github.com/Vidhi123-dev/Global-Fuel-Prices-and-Taxation-Trends/blob/main/project%20tablaue.twbx

Tableau Public dashboard: https://public.tableau.com/authoring/globalfuelprojecttablaue/Dashboard1#1

Contact

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